The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Ground-Generating Substrate Foundation Dynamics
Next Benchmark Criterion

Abstract

The current ground-frontier routing note had fixed the admissible burden order below the theorem forcing the origin-generating substrate ground package $\mathfrak{G}^{\mathrm{sub,grd}}$ from ground-generating substrate foundation dynamics $\mathfrak{F}^{\mathrm{sub,fdn}}$: first try to derive or uniquely force the foundation package from still more primitive explicit substrate structure, and if that could not yet be done cleanly, prove a genuinely smaller theorem-level collapse or sharper necessity / uniqueness result strictly inside foundation-facing scope. The new foundation-forcing theorem has now spent that first admissible route. The ground-generating substrate foundation package $\mathfrak{F}^{\mathrm{sub,fdn}}$ is no longer left as an independently inserted stopping point; it is a canonical and unique output of foundation-generating substrate root dynamics $\mathfrak{R}^{\mathrm{sub,rt}}$.

This note records the routing consequence of that new theorem. The next honest benchmark-facing manuscript must therefore go one level below the current foundation frontier: first derive or uniquely force the foundation-generating substrate root package $\mathfrak{R}^{\mathrm{sub,rt}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status, or, if that cannot be done cleanly, prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the current ground-generating substrate foundation package $\mathfrak{F}^{\mathrm{sub,fdn}}$. Another ground relay, origin-package restatement, precursor-package restatement, lift-package restatement, shell-restriction restatement, split-core restatement, selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, another same-foundation relay that preserves $\mathfrak{F}^{\mathrm{sub,fdn}}$ without shrinking or forcing it further, or another same-root relay that preserves $\mathfrak{R}^{\mathrm{sub,rt}}$ without shrinking or forcing it further does not count next.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^{\sharp}$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and stronger first-principles promotion language remains paused. The positive result of this note is therefore routing discipline at the new foundation frontier.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity is no longer at a stage where the ground-generating substrate foundation package $\mathfrak{F}^{\mathrm{sub,fdn}}$ should be treated as the default unexplained stopping point. The ground-forcing theorem had already shown that the origin-generating substrate ground package $\mathfrak{G}^{\mathrm{sub,grd}}$ is a canonical and unique output of $\mathfrak{F}^{\mathrm{sub,fdn}}$. The new foundation-forcing theorem has now spent the next admissible upstream move below that frontier: $\mathfrak{F}^{\mathrm{sub,fdn}}$ itself is a canonical and unique output of the deeper root class $\mathfrak{R}^{\mathrm{sub,rt}}$.

The present note records the routing consequence of that theorem. It does not reopen the foundation mathematics, and it does not claim a completed first-principles substrate derivation of gravity. It records only which kind of next manuscript would now count as an honest benchmark-facing gain below the current foundation frontier.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It records only which kind of next manuscript would now count as an honest benchmark-facing gain below the foundation frontier.

2. Fixed Inputs

Definition 1 (Foundation-frontier fixed inputs)**.** *The criterion recorded here uses the following fixed inputs.*

1. *The benchmark exact-closure package, which fixes the public one-metric GR target of the repository.*
2. *The benchmark gatekeeping layer, including the post-bridge status correction and the upstream-compression safeguard that blocks benchmark-neutral deeper-origin relays from counting as new front-line progress once the benchmark-relevant object is unchanged.*
3. *The preserved ground-forcing theorem, which already shows that the origin-generating substrate ground package* $\mathfrak{G}^{\mathrm{sub,grd}}$ *is a canonical and unique output of ground-generating substrate foundation dynamics* $\mathfrak{F}^{\mathrm{sub,fdn}}$*.*
4. *The preserved foundation-from-root theorem, which already shows that* $\mathfrak{F}^{\mathrm{sub,fdn}}$ *can be recovered canonically from foundation-generating substrate root dynamics* $\mathfrak{R}^{\mathrm{sub,rt}}$*.*
5. *The new foundation necessity / uniqueness theorem, which proves that* $\mathfrak{F}^{\mathrm{sub,fdn}}$ *itself is a canonical and unique output of* $\mathfrak{R}^{\mathrm{sub,rt}}$*.*

Remark 1. *The present note does not replace those manuscripts. It records the routing consequence of reading them together under the active safeguard against recursive depth drift.*

3. Why the Foundation Theorem Is the Frontier

Definition 2 (Benchmark-neutral foundation relay)**.** *A* benchmark-neutral foundation relay *is a manuscript that preserves the same benchmark one-metric output, the same ordinary matter route, the same foundation package* $\mathfrak{F}^{\mathrm{sub,fdn}}$*, the same ground package* $\mathfrak{G}^{\mathrm{sub,grd}}$*, the same origin*

package $\mathfrak{O}^{\mathrm{sub,orig}}$*, the same precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$*, the same lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*, the same shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*, the same split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, the same selector-law consequence* $\mathfrak{S}^{\mathrm{sel,0}}$*, the same combined-response consequence* $\mathfrak{R}^{\mathrm{comb}}$*, the same ambient-package consequence* $\mathfrak{A}^{\mathrm{bal}}$*, the same trace-core consequence* $\mathfrak{C}^{\mathrm{tr}}$*, the same completion-core consequence* $\mathfrak{C}^{\mathrm{zr,aff}}$*, the same exact-shell affine nucleus consequence* $\mathfrak{N}^{\mathrm{aff}}$*, and the same claim boundary while merely restating* $\mathfrak{F}^{\mathrm{sub,fdn}}$*,* $\mathfrak{G}^{\mathrm{sub,grd}}$*,* $\mathfrak{O}^{\mathrm{sub,orig}}$*,* $\mathfrak{P}^{\mathrm{sub,prec}}$*,* $\mathfrak{L}^{\mathrm{sub,lift}}$*,* $\mathfrak{S}^{\mathrm{sub,sh}}$*,* $\mathfrak{C}^{\mathrm{sel,split}}$*,* $\mathfrak{S}^{\mathrm{sel,0}}$*,* $\mathfrak{R}^{\mathrm{comb}}$*,* $\mathfrak{A}^{\mathrm{bal}}$*,* $\mathfrak{C}^{\mathrm{tr}}$*,* $\mathfrak{C}^{\mathrm{zr,aff}}$*,* $\mathfrak{N}^{\mathrm{aff}}$*, or the same root consequence without shrinking or forcing the foundation or root package further.*

Proposition 1 (The foundation-forcing theorem is the live frontier)**.** *At the current stage of the primitive-reservoir observer-reduction line, the theorem forcing the ground-generating substrate foundation package* $\mathfrak{F}^{\mathrm{sub,fdn}}$ *from foundation-generating substrate root dynamics* $\mathfrak{R}^{\mathrm{sub,rt}}$ *is the live benchmark-facing frontier.*

Proof. That theorem changes benchmark-facing status at current scope because it removes the foundation package $\mathfrak{F}^{\mathrm{sub,fdn}}$ itself as the live internal stopping point. The current bridge datum is no longer merely a deeper package whose ground consequence was already known; it is now forced from the root layer. By contrast, a manuscript that preserves the same benchmark package, the same foundation consequence, and the same ground/origin/precursor/lift/shell-restriction consequences while merely restating the foundation package or another already-forced larger or smaller bridge object does not alter benchmark-facing status unless it adds a comparably sharp gain. Therefore the live frontier is the foundation-forcing theorem rather than the earlier ground-forcing theorem or a later same-foundation relay. □

Corollary 1 (Status of the foundation package as a next-step target)**.** *Under the current routing safeguard, the ground-generating substrate foundation package* $\mathfrak{F}^{\mathrm{sub,fdn}}$ *is no longer itself the default next benchmark-facing target.*

Proof. The foundation-forcing theorem already proves that $\mathfrak{F}^{\mathrm{sub,fdn}}$ is a canonical and unique output of $\mathfrak{R}^{\mathrm{sub,rt}}$. Once that is on record, another manuscript that spends the move on the foundation package again does not remove any further benchmark-relevant freedom. The open burden has therefore moved below foundation scope. □

4. The Next Benchmark Criterion

Definition 3 (Admissible next benchmark-facing foundation manuscript)**.** *At the current foundation frontier, a manuscript counts as an admissible next benchmark-facing move only if it respects the following burden order:*

1. *its primary attempt is to derive or uniquely force the foundation-generating substrate root package* $\mathfrak{R}^{\mathrm{sub,rt}}$ *from still more primitive explicit substrate structure in a way that does more than merely relocate the unexplained layer deeper;*

2. *if that cannot be done cleanly at current scope, its admissible fallback gain is to prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the ground-generating substrate foundation package* $\mathfrak{F}^{\mathrm{sub,fdn}}$*;*

3. *in either case it preserves the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary, and it introduces neither a second operative metric nor an enlarged low-energy matter law and licenses no stronger first-principles promotion language.*

Theorem 1 (Next honest benchmark-facing task at foundation scope)**.** *The next honest benchmark-facing manuscript at the current foundation frontier should therefore go one level below that frontier: first try to derive or uniquely force the foundation-generating substrate root package* $\mathfrak{R}^{\mathrm{sub,rt}}$ *from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status. If that cannot be done cleanly, the admissible fallback is to prove a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside the ground-generating substrate foundation package* $\mathfrak{F}^{\mathrm{sub,fdn}}$*. A manuscript that only restates the origin-generating substrate ground package* $\mathfrak{G}^{\mathrm{sub,grd}}$*, only restates the precursor-generating substrate origin package* $\mathfrak{O}^{\mathrm{sub,orig}}$*, only restates the lift-generating substrate precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$*, only restates the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*, only restates the shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*, only restates the split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, only restates the selector law* $\mathfrak{S}^{\mathrm{sel},0}$*, only restates the combined response law* $\mathfrak{R}^{\mathrm{comb}}$*, only restates the ambient package* $\mathfrak{A}^{\mathrm{bal}}$*, only restates the trace core* $\mathfrak{C}^{\mathrm{tr}}$*, only restates the completion-core presentation* $\mathfrak{C}^{\mathrm{zr,aff}}$*, only restates the exact-shell affine nucleus* $\mathfrak{N}^{\mathrm{aff}}$*, only repeats the same foundation consequence while preserving* $\mathfrak{F}^{\mathrm{sub,fdn}}$ *without shrinking or forcing it further, or only repeats the same root consequence while preserving* $\mathfrak{R}^{\mathrm{sub,rt}}$ *without shrinking or forcing it further does not satisfy the criterion.*

Proof. By Proposition 1, the live frontier already lies at the theorem that forces the foundation package from root dynamics. Therefore a second manuscript below it must improve on that status rather than merely accompany it. A deeper-origin manuscript can do so only if it changes benchmark-facing status by more than depth alone, and the first clean way to do that is to force $\mathfrak{R}^{\mathrm{sub,rt}}$ itself from still more primitive explicit substrate structure. If that first option is not yet available cleanly, the routing safeguard allows the fallback of a smaller internal collapse, necessity result, or uniqueness result inside $\mathfrak{F}^{\mathrm{sub,fdn}}$. If a manuscript simply reproduces the same foundation consequence through another relay while preserving the same foundation or root package without shrinking or forcing it further, the routing rule classifies it as benchmark neutral. □

5. What the Next Move Should Not Be

Proposition 2 (Screened next moves)**.** *At the current foundation frontier, none of the following is the default next benchmark-facing move:*

1. *another restatement of the origin-generating substrate ground package* $\mathfrak{G}^{\mathrm{sub,grd}}$*;*
2. *another restatement of the precursor-generating substrate origin package* $\mathfrak{O}^{\mathrm{sub,orig}}$*;*
3. *another restatement of the lift-generating substrate precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$*;*
4. *another restatement of the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*;*
5. *another restatement of the primitive substrate shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*;*
6. *another restatement of the split zero-hidden selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*;*
7. *another restatement of the full zero-hidden selector law* $\mathfrak{S}^{\mathrm{sel},0}$*;*

8. *another restatement of the combined response substrate law* $\mathfrak{R}^{\mathrm{comb}}$*;*
9. *another restatement of the completion-balance ambient dynamics package* $\mathfrak{A}^{\mathrm{bal}}$*;*
10. *another restatement of the completion-state shell-trace core* $\mathfrak{C}^{\mathrm{tr}}$*;*
11. *another restatement of the full completion-core presentation* $\mathfrak{C}^{\mathrm{zr,aff}}$*;*
12. *another restatement of the exact-shell affine nucleus* $\mathfrak{N}^{\mathrm{aff}}$*;*
13. *another foundation relay that preserves the same* $\mathfrak{F}^{\mathrm{sub,fdn}}$ *without shrinking or forcing it further;*
14. *another root relay that preserves the same* $\mathfrak{R}^{\mathrm{sub,rt}}$ *without shrinking or forcing it further;*
15. *a manuscript that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger first-principles promotion language.*

Proof. Items (1)–(14) fail the preceding criterion because they do not directly address the current root-facing burden. They revisit either already-forced larger bridge objects, already-forced smaller bridge objects, the same foundation stopping point, or the same root stopping point rather than removing the remaining root freedom or proving a still smaller internal datum inside $\mathfrak{F}^{\mathrm{sub,fdn}}$. Item (15) violates the fixed claim boundary of the benchmark package and therefore cannot count as a disciplined next step on the active line. The benchmark package remains the exact one-metric GR package already fixed by adoption, so any admissible next manuscript must preserve that boundary while keeping stronger first-principles promotion language paused. □

Remark 2. *This screening statement is not a denial that the ground-forcing theorem, the origin-forcing theorem, the precursor-forcing theorem, the lift-forcing theorem, the shell-restriction forcing theorem, the selector-law collapse theorem, the selector-law forcing theorem, the combined-response theorem, the ambient-package theorem, the trace-core theorem, or the completion-core theorem matter historically. It is a routing statement: they are not the honest way to spend the next benchmark-facing move from the current foundation frontier.*

6. Conclusion

The positive result of this note is routing discipline at the current foundation frontier. The foundation-forcing theorem remains the live benchmark-facing gain because it already removes the ground-generating substrate foundation package as the internal stopping point on the active line. The foundation package therefore no longer sets the honest next burden, and neither do the already-forced ground package, origin package, precursor package, lift package, shell-restriction package, split core, selector law, combined response law, ambient package, trace core, completion-core presentation, or exact-shell affine nucleus.

The scope remains narrow and honest. The benchmark package is unchanged: the one operative metric is still exactly $g^{\sharp}$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to derivation or licenses stronger first-principles promotion language yet.

The next open burden is therefore clear. The next benchmark-facing manuscript should go one level below the current foundation frontier: first try to derive or uniquely force the foundation-generating substrate root package $\mathfrak{R}^{\mathrm{sub,rt}}$ from still more primitive explicit substrate structure in a

way that actually changes benchmark-facing status while preserving the same benchmark one-metric package and claim boundary. If that cannot be done cleanly, the admissible fallback is a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside the ground-generating substrate foundation package $\mathfrak{F}^{\mathrm{sub,fdn}}$. Another ground relay, origin relay, precursor relay, lift relay, shell-restriction relay, split-core restatement, selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, same-foundation relay, or same-root relay that preserves the current packages without shrinking or forcing them further is not the clean next manuscript target at current scope.